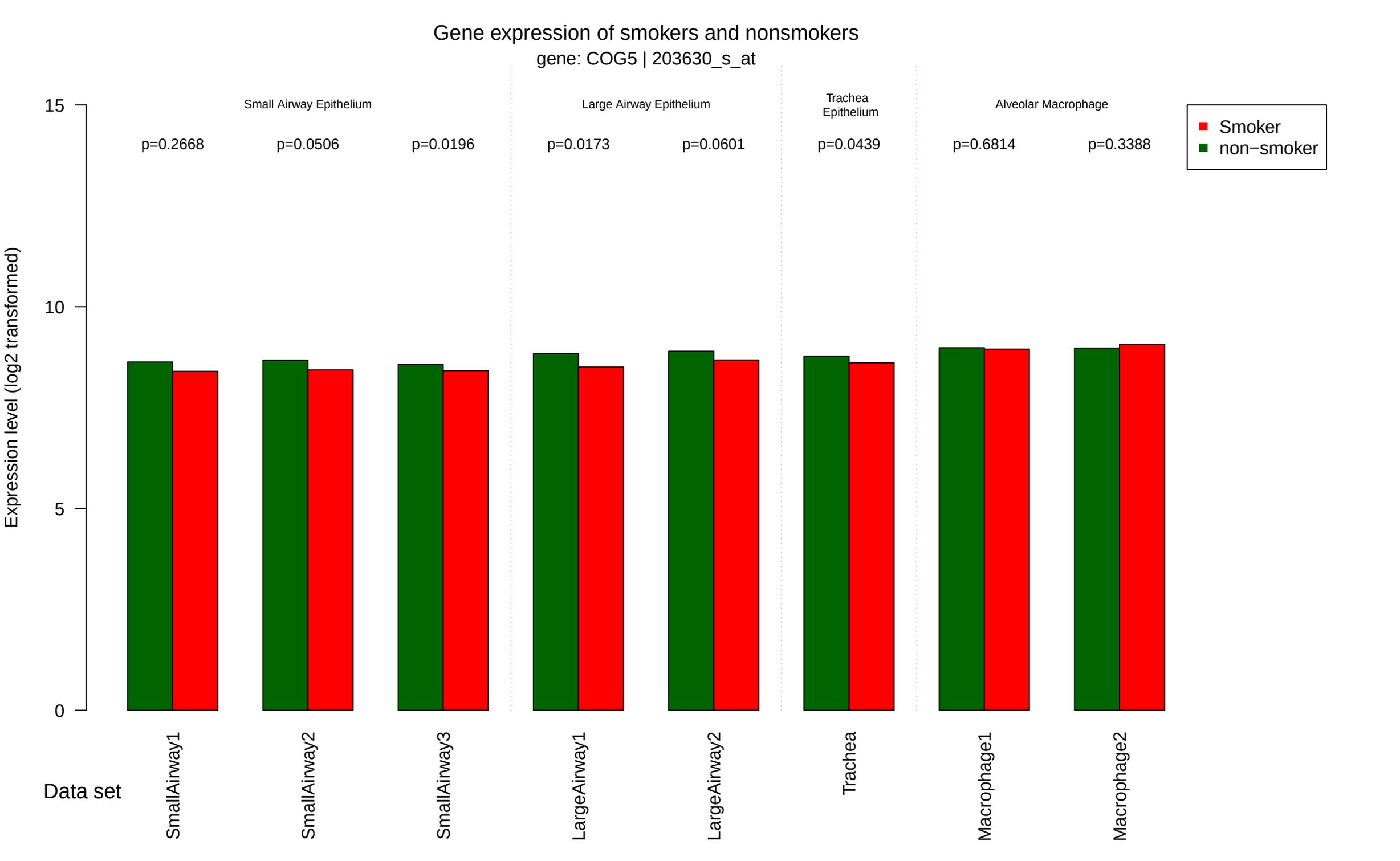
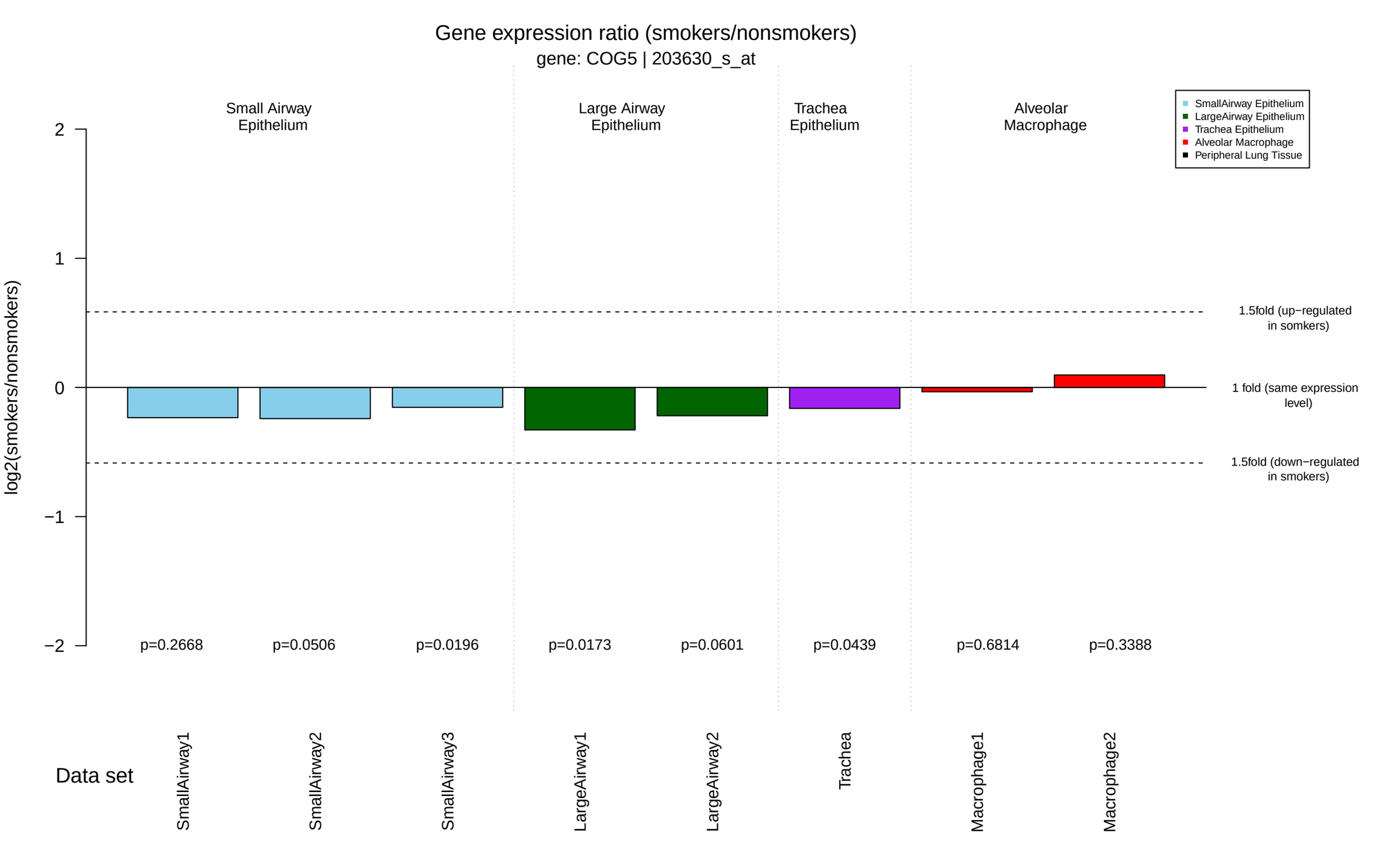
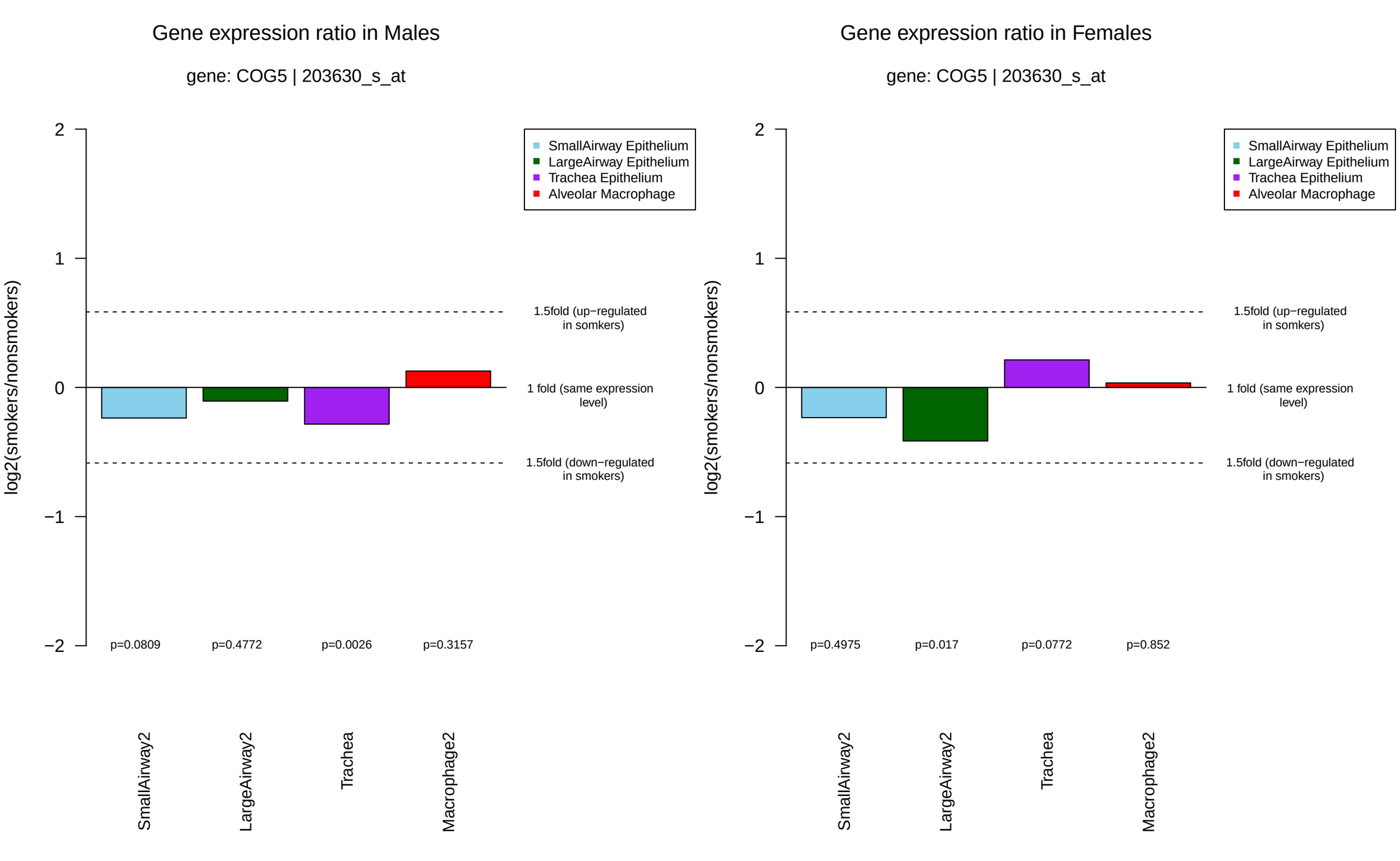
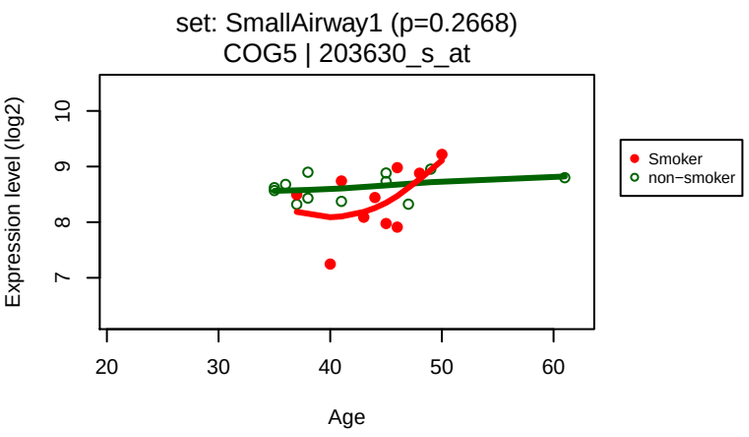
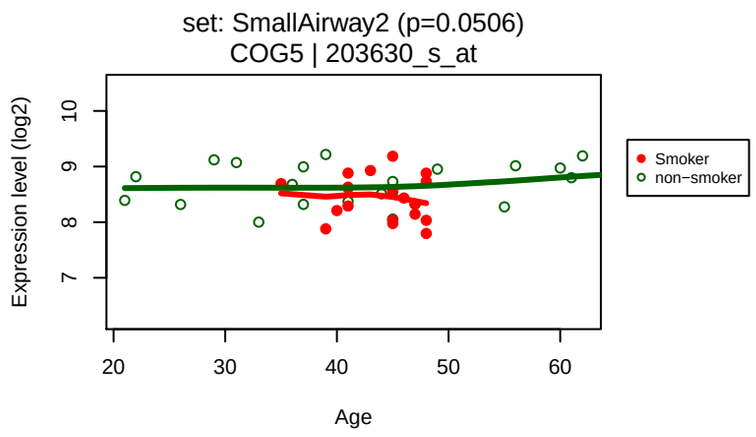
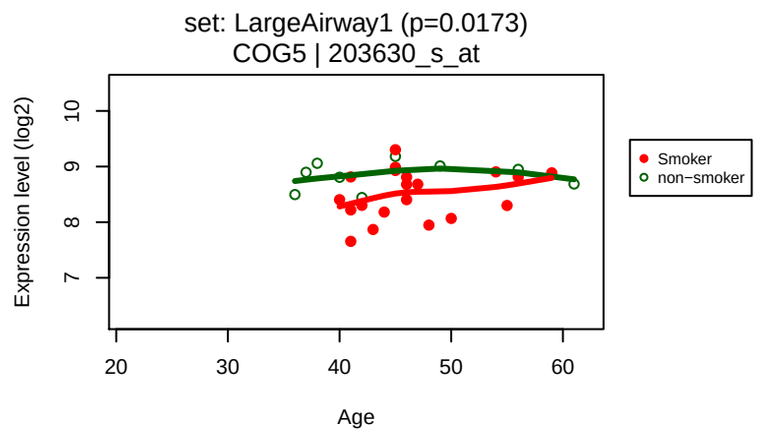

Gene expression of smokers and nonsmokers
gene: COG5 | 203630_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2668
p=0.0506
p=0.0196
p=0.0173
p=0.0601
p=0.0439
p=0.6814
p=0.3388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

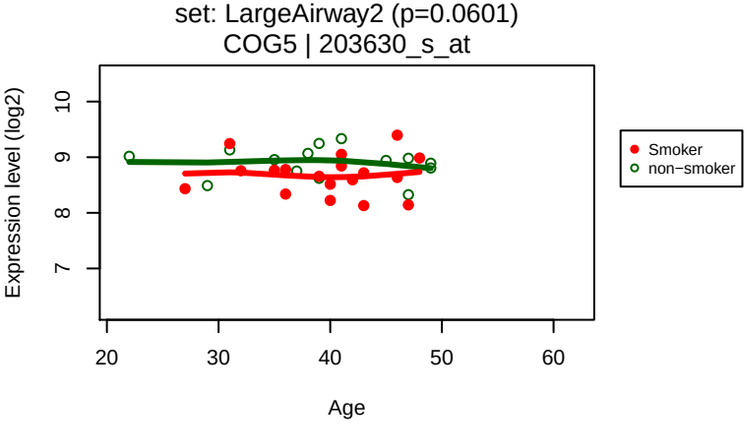

Gene expression ratio (smokers/nonsmokers)
gene: COG5 | 203630_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2668
p=0.0506
p=0.0196
p=0.0173
p=0.0601
p=0.0439
p=0.6814
p=0.3388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

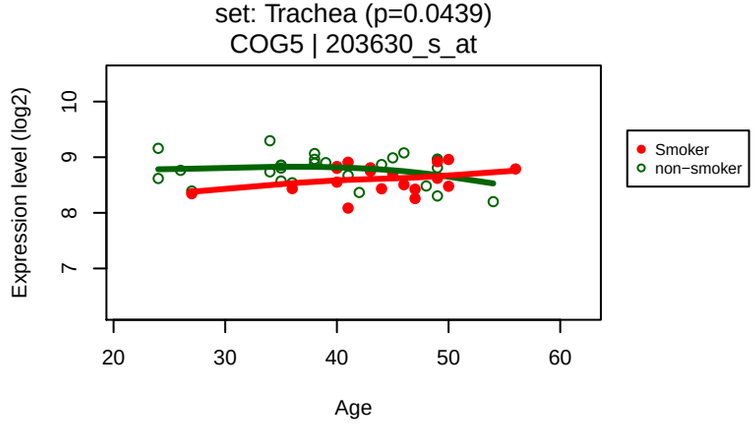

Gene expression ratio in Males
gene: COG5 | 203630_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0809
p=0.4772
p=0.0026
p=0.3157
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COG5 | 203630_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4975
p=0.017
p=0.0772
p=0.852
SmallAirway2
LargeAirway2
Trachea
Macrophage2

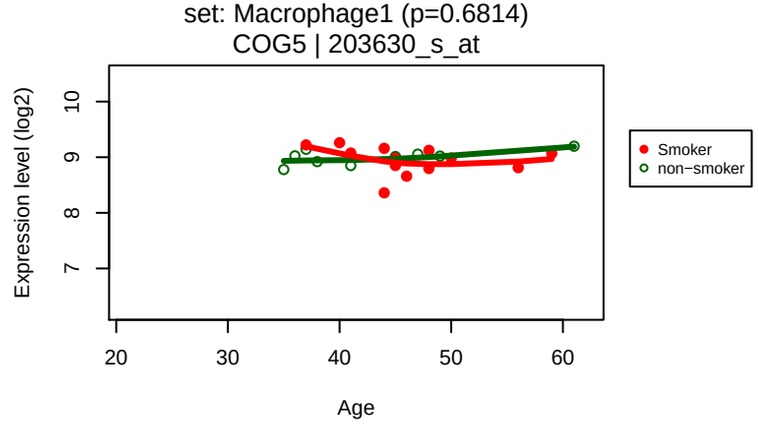

set: SmallAirway1 (p=0.2668)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker

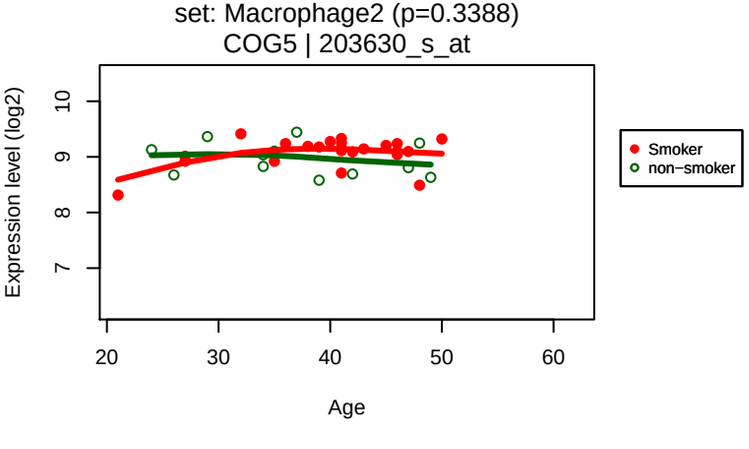

set: SmallAirway2 (p=0.0506)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker

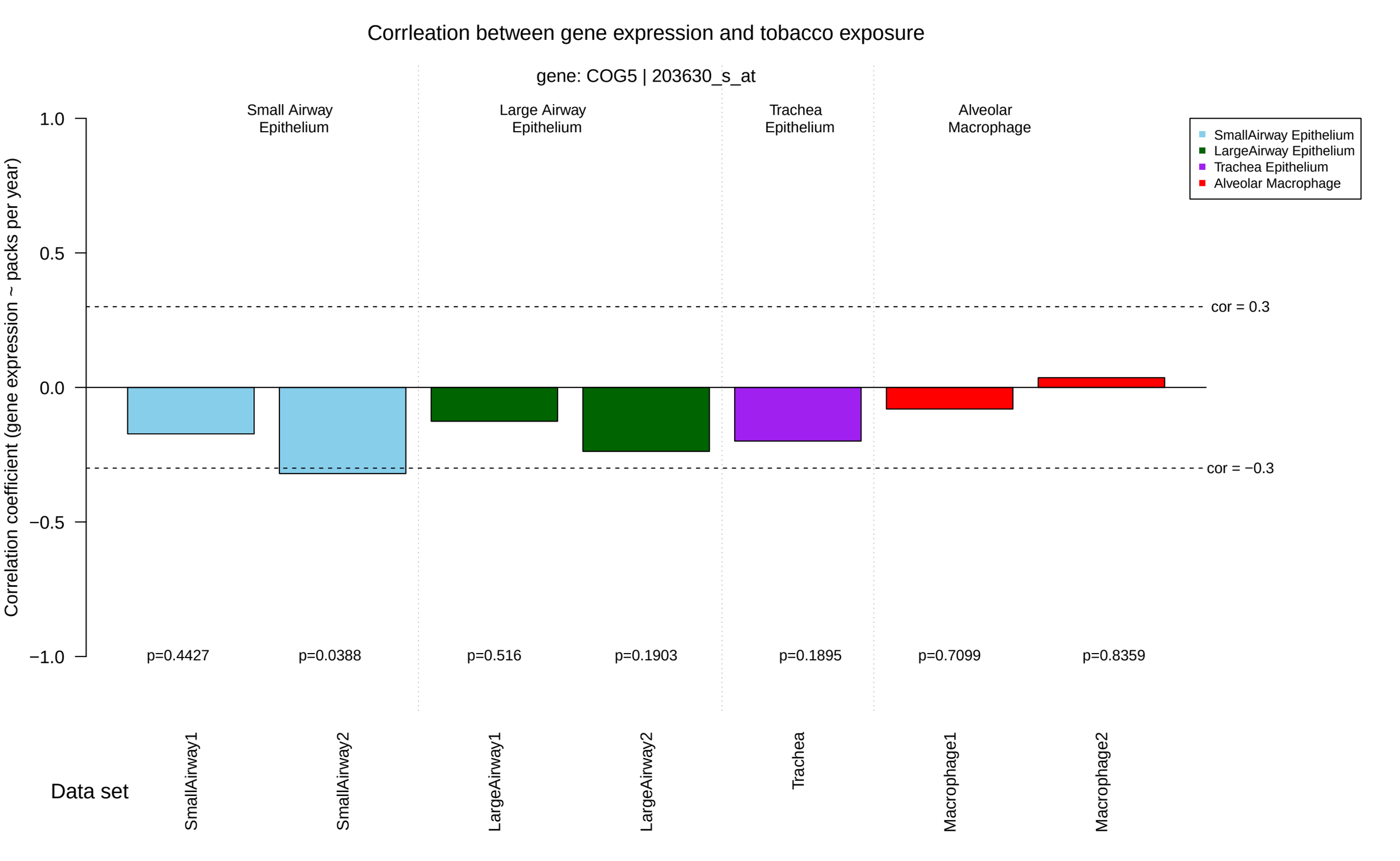

set: LargeAirway1 (p=0.0173)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0601)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0439)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6814)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3388)
COG5 | 203630_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COG5 | 203630_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4427
p=0.0388
p=0.516
p=0.1903
p=0.1895
p=0.7099
p=0.8359
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2